



Department of Electrical and Electronics Engineering

Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: V Semester B.E. EEE

Course Code & Title: EE8501 Power System Analysis

Name of the Faculty member (s): Mr. A. Arun Kumar

Innovative Practice Description

- **Unit / Topic:** Unit I / Y-bus formation-Singular transformation methods
- **Course Outcome:** CO 1
- **Topic Learning Outcome:** TLO 3
- **Activity Chosen:** One minute paper
- **Justification:**

The Y_{BUS} can be assembled by the use of singular transformations given by a graph theoretical approach. Since, one minute paper activity provides a conceptual bridge between successive class periods. Improve the quality of class discussion by having students write briefly about a concept.

- **Time Allotted for the Activity:** 5 minutes
- **Details of the Implementation:**

At the end of the class, students were asked to write about the topic discussed in the class. The students expressed the understood content and the content which were not clear in that particular topic. This activity shows whether the students can able to understand the specific topic and their involvement the particular class.

- **CO – PO / PSO mapping:**

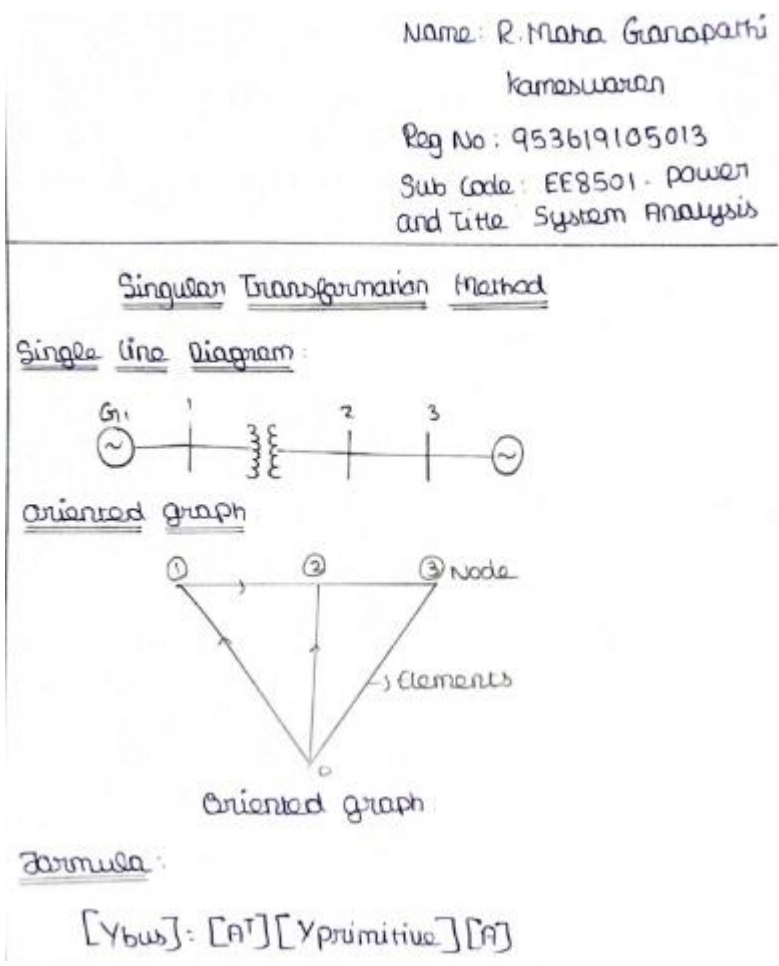
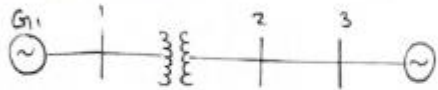
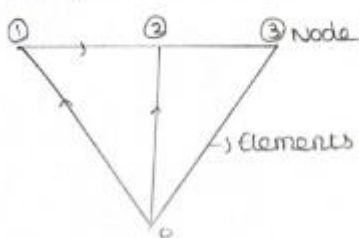
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO 1	3	2	1	1	1	-	1	-	-	-	-	1	3	-	-

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO9
	1
Justification for correlation	The students can Function effectively as an individual

- **Images / Screenshot of the practice:**

Innovative Teaching Method Execution
Y-bus formation-Singular transformation methods
– One minute paper
 Name: R. Maha Ganapathi kameswaran Reg No : 953619105013 Sub Code : EE8501 - power and Title System Analysis <u>Singular Transformation Method</u> <u>Single line Diagram:</u>  <u>Oriented graph</u>  <u>Formula:</u> $[Y_{bus}] = [A^T][Y_{primitive}][A]$

- **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- ✓ Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
 - ✓ Students can able to attend the question even in the questions are in indirect form.
 - ✓ Students can able to explain the concepts in examination without any confusion.
- ❖ ***Challenges faced in implementation:***
 - ✓ Time utilization for conducting activity.

References:

1. John J. Grainger, William D. Stevenson, Jr, 'Power System Analysis', Mc Graw Hill Education (India) Private Limited, New Delhi, 2015.
2. Kothari D.P. and Nagrath I.J., 'Power System Engineering', Tata McGraw-Hill Education, Second Edition, 2008.
3. Hadi Saadat, 'Power System Analysis', Tata McGraw Hill Education Pvt. Ltd., New Delhi, 21st reprint, 2010.

Signature of Faculty Member

HOD



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Name of the Faculty member (s): Mr. A. Arun Kumar

Innovative Practice Description

- **Unit / Topic:** Unit III / Flow chart and Problems in NR method in Newton-Raphson (NR) method

- **Course Outcome:** CO 2

- **Topic Learning Outcome:** TLO 6

- **Activity Chosen:** Concept map

- **Justification:**

Concept mapping is a creative way to set goals, solve problems and design action plans. It quickly records ideas in a free-form way. When groups use mini mapping, the thoughts of each participant easily trigger ideas in others. This dynamic group interaction encourages breaking free of old patterns to uncover new and innovative approaches. For this topic, there are different types of magnetic material and its characteristics can be brought in to single map for better understanding.

- **Time Allotted for the Activity:** 10 minutes

- **Details of the Implementation:**

Mini mapping activities require students to actively engage in their learning, often by connecting their prior knowledge to new information. When creating a mini map, a student frequently interacts with a textbook, notes from class, an instructor, classmate, or study group.

- In my subject Mini map was conducted for the topic Magnetic materials – Magnetization. The students are voluntarily drawn the map.

- **CO – PO / PSO mapping:**

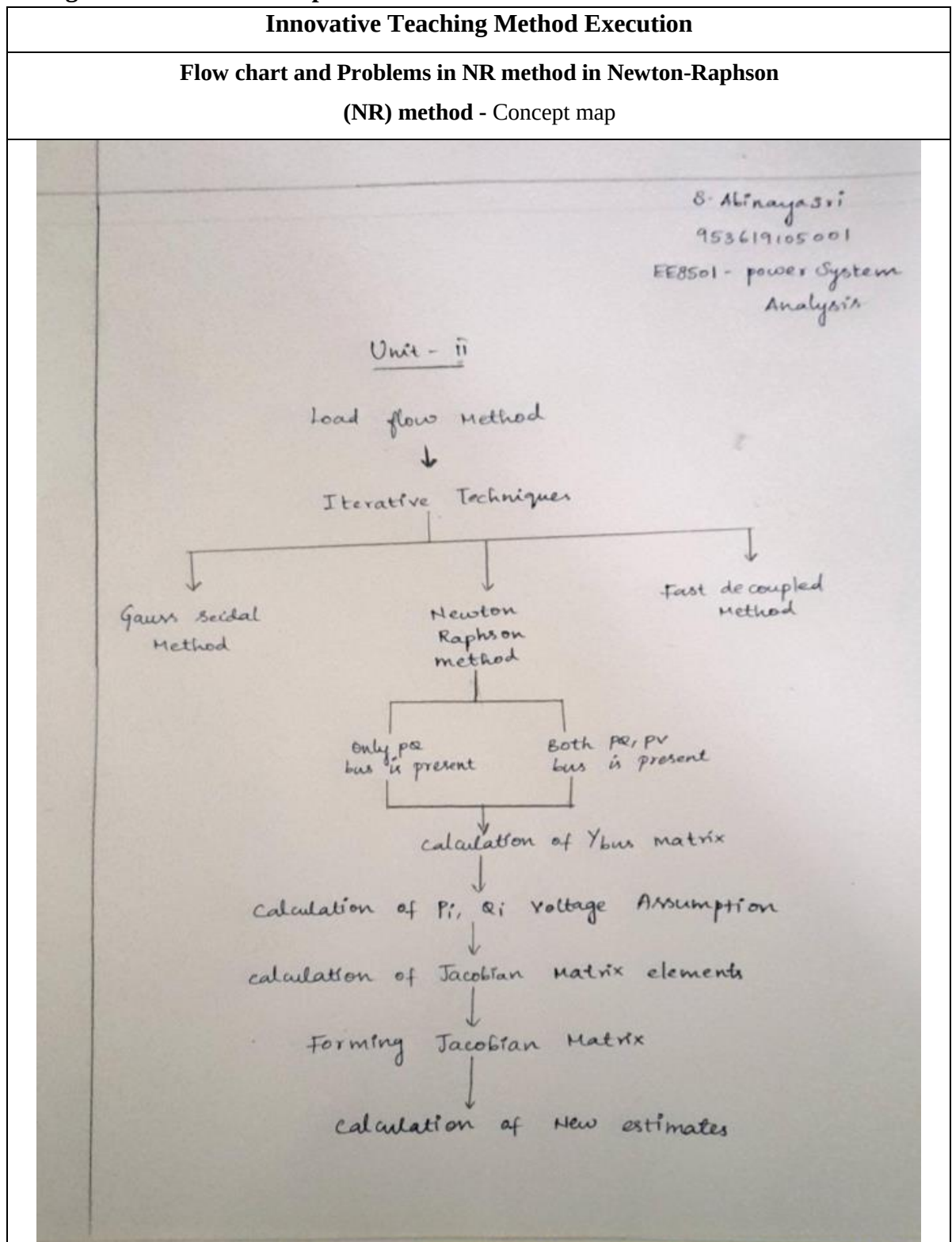
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO2	3	1	1	2	1	-	1	-	-	-	-	1	3	-	-

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO9
	1
Justification for correlation	The students can Function effectively as an individual

- Images / Screenshot of the practice:



- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The benefits of Mini Maps are a great way for students to make notes on all of the information they receive. It helps the students to note down only the most important information using key words, and then make connections between facts and ideas visually – keeping all of your topic thoughts together on one sheet. It made key note making easier to students, as it reduces pages of notes into one single side of paper. Also mini map made slow learners to remember the information more quickly.

Challenges faced in implementation:

Normally teachers will give longer explanations in the notes section of the topic. The students are made into groups and to draw the map to indicate relationships between the topics in biomass conversion process. I planned the activity for 10 minutes only. But in real scenario it takes 15 minutes to complete this activity.

References:

1. John J. Grainger, William D. Stevenson, Jr, 'Power System Analysis', Mc Graw Hill Education (India) Private Limited, New Delhi, 2015.
2. Kothari D.P. and Nagrath I.J., 'Power System Engineering', Tata McGraw-Hill Education, Second Edition, 2008.
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Name of the Faculty member (s): Mr. A. Arun Kumar

Innovative Practice Description

- **Unit / Topic:** Unit III / Methods to solve short circuit analysis and Symmetrical short circuit analysis using Thevenin's theorem
- **Course Outcome:** CO 3
- **Topic Learning Outcome:** TLO 8
- **Activity Chosen:** Simulation
- **Justification:**

The uncertainty in power system causes huge losses. One such uncertainty is Fault (Symmetrical and Unsymmetrical fault). The fault cannot be experimented in real time to understand the effect. It is expected that, the students will have better understanding after the usage of simulation software for this topic.

Time Allotted for the Activity: 20 minutes

- **Details of the Implementation:**

The students were asked to design and analyses the cause and effect of fault analysis effectively using the Power World Simulator.

CO – PO / PSO mapping:

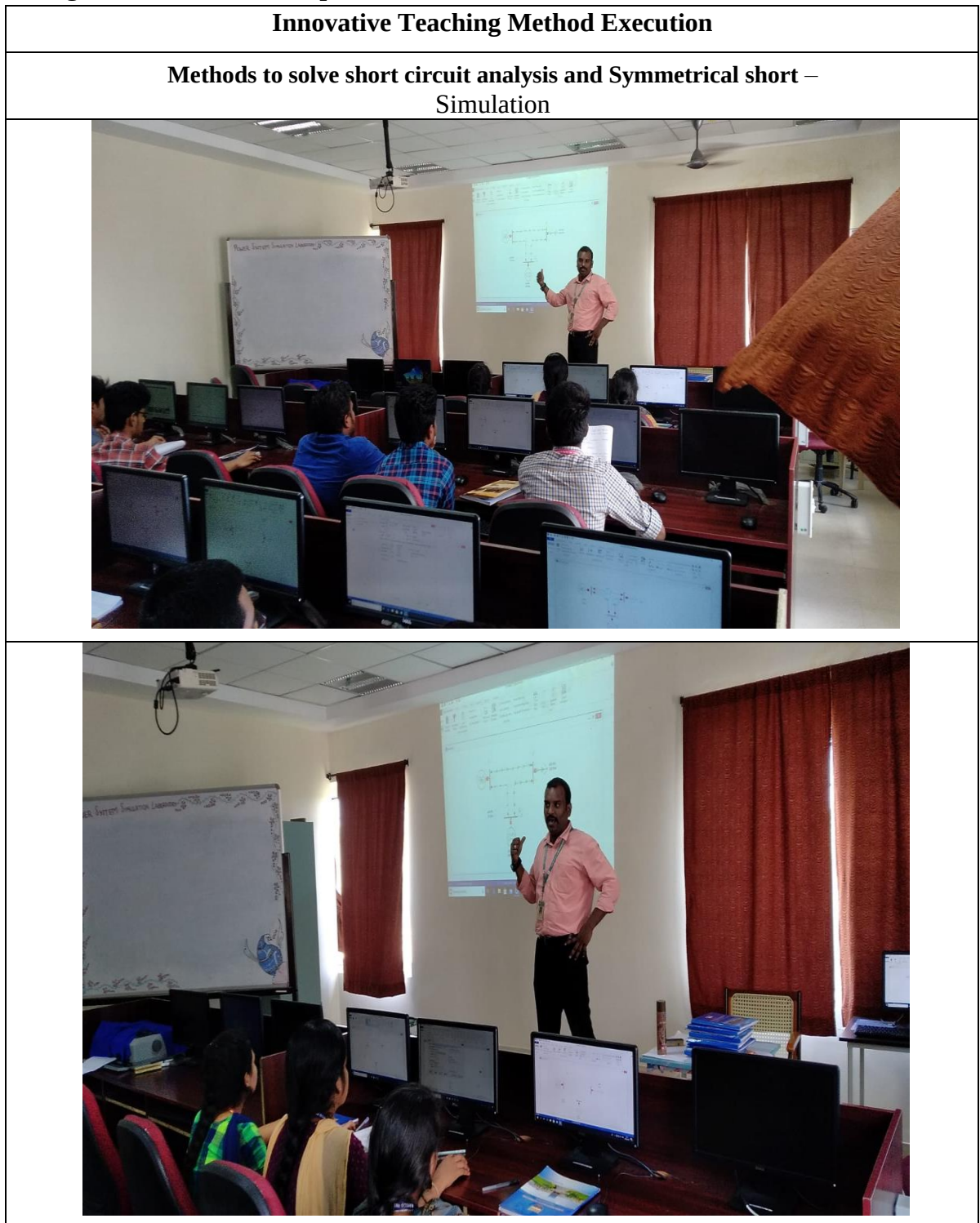
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO5	3	3	1	2	1	-	1	-	-	-	-	1	3	-	-

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO5
	1
Justification for correlation	The students can effectively use the simulation software tool.

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

- ✓ Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.
 - ✓ Also they can able to analyse the real time problem.

- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The process of simulating a scenario to practice different responses and actions to a real life situation is extremely effective in knowledge retention. This is because knowledge isn't in theory – the user needs to apply it in a real-life situation. Simulations can be slowed down to study behavior more closely. Conditions can be varied and outcomes investigated. Also Critical situations can be investigated without risk.

- ❖ ***Challenges faced in implementation:***

I planned the activity for 20 minutes. But in Class room it takes 30 minutes.

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1. John J. Grainger, William D. Stevenson, Jr, 'Power System Analysis', Mc Graw Hill Education (India) Private Limited, New Delhi, 2015.
2. Kothari D.P. and Nagrath I.J., 'Power System Engineering', Tata McGraw-Hill Education, Second Edition, 2008.
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Innovative Practice Description

- **Unit / Topic:** Unit IV / Computation of post fault currents in symmetrical component and phasor domains
- **Course Outcome:** CO 4, CO 6
- **Topic Learning Outcome:** TLO 12
- **Activity Chosen:** Simulation
- **Justification:**

The uncertainty in power system causes huge losses. One such uncertainty is Fault (Symmetrical and Unsymmetrical fault). The fault cannot be experimented in real time to understand the effect. It is expected that, the students will have better understanding after the usage of simulation software for the topic computation of post fault currents in symmetrical component and phasor domains.

Time Allotted for the Activity: 20 minutes

- **Details of the Implementation:**

The students were asked to design and analyses the post fault currents in symmetrical fault analysis effectively using the Power World Simulator.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO4	3	3	1	2	1	-	1	-	-	-	-	1	3	-	-
CO6	3	3	1	1	1	-	1	-	-	-	-	1	3	-	-

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO5
	1
Justification for correlation	The students can effectively use the simulation software tool.

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- ✓ Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.
- ✓ Also they can able to analyse the real time problem.

- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The process of simulating a scenario to practice different responses and actions to a real life situation is extremely effective in knowledge retention. This is because knowledge isn't in theory – the user needs to apply it in a real-life situation. Simulations can be slowed down to study behavior more closely. Conditions can be varied and outcomes investigated. Also Critical situations can be investigated without risk.

- ❖ ***Challenges faced in implementation:***

I planned the activity for 20 minutes. But in Class room it takes 30 minutes.

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Innovative Practice Description

- **Unit / Topic:** Unit V / Classical step-by-step solution of the swing equation
- **Course Outcome:** CO 5
- **Topic Learning Outcome:** TLO 15
- **Activity Chosen:** Think pair share
- **Justification:**
 - ✓ It helps students to think individually about a topic or answer to a question.
 - ✓ It teaches students to share ideas with classmates and builds oral communication skills.
 - ✓ It helps focus attention and engage students in comprehending the reading material.
- **Time Allotted for the Activity:** 10 minutes
- **Details of the Implementation:**

Think-Pair-Share innovative practice conducted for the concept of classical step-by-step solution of the swing equation. First, I asked the students to think about the different elements for 2 minutes. Then I make them as a pair to discuss their neighbour's and asked the students to discuss about the equation for different elements for 3 minutes. Finally I asked the one of the team to explain the concept to whole class for further discussion. The students from group share their points and participated in the discussion for 10 minutes.

• CO – PO / PSO mapping:

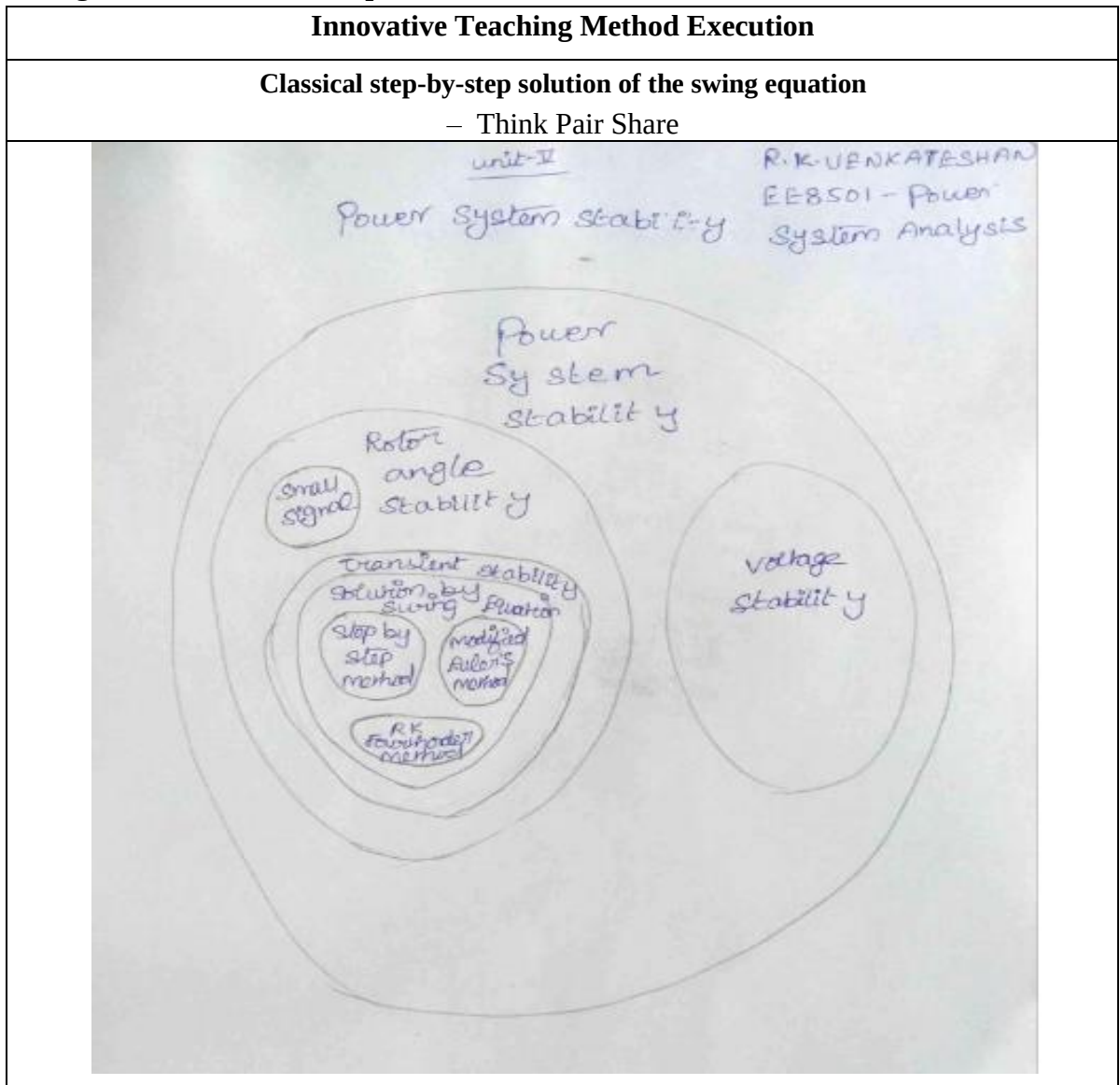
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CO5	3	3	1	1	1	-	1	-	-	-	-	1	3	-	-

(1 – Low 2 – Moderate 3 – High)

• PO / PSO mapped:

Innovative practice	PO9
	1
Justification for correlation	The students can Function effectively as a team

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

- ✓ Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

- ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

Think-pair-sharing forces all students to attempt an initial response to the question, which they can then clarify and expand as they collaborate. It also gives them a chance to validate their ideas in a small group before mentioning them to the large group, which may help shy students feel more confident participating.

- ❖ **Challenges faced in implementation:**

I planned the activity for 10 minutes. But in Class room it takes 15 minutes.

References:

1. John J. Grainger, William D. Stevenson, Jr, 'Power System Analysis', Mc Graw Hill Education (India) Private Limited, New Delhi, 2015.
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